Evan Bell

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Education

Massachusetts Institute of Technology May 2025
S.B. Mathematics; S.B. Computer Science and Engineering *Cambridge, MA*
GPA: 4.8/5.0
Coursework: Deep Learning, Machine Learning, NLP, Probability, Real Analysis, Algorithms

Professional Experience

Google – YouTube Aug 2025 – Present
Software Engineer *San Bruno, CA*

- Designed and implemented C++ backend ranking and selection components, integrating revenue signals and rule-based prioritization through production RPC interfaces.
- Built an offline data reconciliation pipeline using Flume and Spanner to detect and repair inconsistencies in user data records.
- Owned backend component delivery from design and RPC integration through deployment and production debugging; supported releases across three international markets.

SESCO Enterprises Summer 2025
Software Engineering / Quantitative Research Intern

- Optimized numerical routines in electricity-grid modeling software, targeting runtime and cache-efficient memory access; evaluated alternative algorithms for forecasting and simulation.
- Researched and designed a Rust architecture to replace Python market-data parsing, evaluating parsing efficiency and processing overhead.

Dimensional Insight Summer 2024
Software Engineering Intern

- Optimized C-based retrieval over large data files, reducing recurring-query processing time from approximately five minutes to seconds.
- Implemented C/C++ encryption functionality to protect sensitive data stored on customer systems.

General Motors – Corvette Summer 2023
Software Engineering Intern

- Developed manufacturing software, drive-load sharing logic, and production-floor diagnostic interfaces for Corvette production; work was projected to reduce delays by 7% in the affected process.

Selected Technical Projects

Quoridor – AlphaZero-Inspired Agent *Python, PyTorch, Numba*

- Co-developed a game-playing agent combining Monte Carlo Tree Search, neural policy/value estimation, and self-play training with legal-action masking.
- Optimized the game engine to approximately 50,000 moves/second on a MacBook using Numba JIT, NumPy-based game-state representations, and hot-path optimization.
- Experimented with policy/value network architectures and Monte Carlo Tree Search budgets.

MIT PokerBots – Heads-Up Poker Agent *Python, C++*

- Developed K-means abstractions of hand and flop states and a strategy inspired by counterfactual regret minimization (CFR) for heads-up poker.

Technical Skills

Languages: C++, Python, C, Julia
ML / Scientific Computing: PyTorch, TensorFlow, NumPy, Numba
Systems / Tools: Flume, Spanner, RPCs, Git, Linux, Valgrind

Activities

MIT Division I Varsity Heavyweight Crew; MIT Poker Club